A permutation algorithm for adjusting the correlation between two variables while preserving their marginal distributions

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Abstract

test

1 Introduction

For various purposes, inducing particular dependence in data. Similarities to datasaurus dozen Matejka and Fitzmaurice (2017), inverse of a permutation test for a correlation (DiCiccio and Romano, 2017), and some kind of empirical approximation to copulas (Schmidt, 2007). Randomized discrete gradient descent.

2 Derivation of the algorithm

The Pearson correlation coefficient ρ_{XY} of two random variables X, Y is defined as the covariance scaled by their standard deviations:

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\sqrt{\mathbb{E}\left[X^2\right] - \mathbb{E}[X]^2}\sqrt{\mathbb{E}\left[Y^2\right] - \mathbb{E}[Y]^2}} \tag{1}$$

This correlation coefficient ρ_{XY} can be estimated from a sample of these random variables — the vectors $\mathbf{x}, \mathbf{y}$. It is common to call the estimate r_{XY}. Assuming the vectors have first been centered (i.e., $\bar{\mathbf{x}} = \bar{\mathbf{y}} = 0$), the estimate is a function of inner products only:

$$r_{XY} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}\sqrt{\langle \mathbf{y}, \mathbf{y} \rangle}} \tag{2}$$

This paper defines an iterative algorithm which permutes values of $\mathbf{y}$ to obtain a target sample correlation ϱ_{XY} or, equivalently, a target raw inner product value

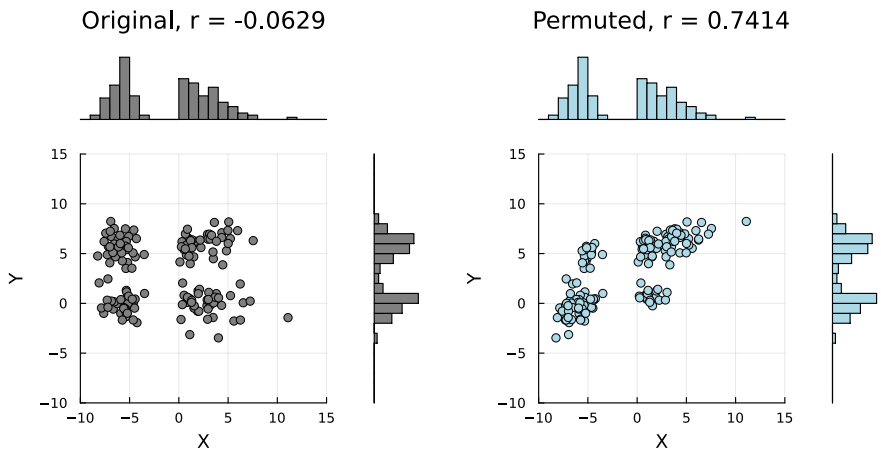

Figure 1: Permuting a random draw of 152 datapoints from two independent mixture distributions: a Normal-Chisquare (X) and a Laplace-Normal mixture (Y). The left panel shows the original random draw, with a sample correlation of -0.0629; the right panel shows the data after permutation with our algorithm, with the target correlation set to 0.7414. The target correlation was achieved to 4 decimals in 65 iterations. Note that the marginal distributions remain exactly equal.

$$\tau_{XY} = \varrho_{XY} \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} \sqrt{\langle \mathbf{y}, \mathbf{y} \rangle} \tag{3}$$

By relying on permutation rather than transformation of the data, we ensure that the marginal distribution of each vector is completely unchanged. An example of the algorithm is shown in Figure 2.

3 Permutation algorithm

Change in the inner product $\langle x, y \rangle$ when permuting two elements of y to y' is derived as follows. Assume, without loss of generality, that $y = y_1, y_2, ..., y_i, y_j, ...y_N$ and $y' = y_1, y_2, ..., y_j, y_i, ..., y_N$.

Then, the change in inner product is as follows:

$$\begin{aligned}
\Delta\tau &= \langle x, y \rangle - \langle x, y' \rangle \\
&= (x_1 y_1 + x_2 y_2 + ... + x_i y_i + x_j y_j + ... + x_N y_N) - \\
&\quad (x_1 y_1 + x_2 y_2 + ... + x_i y_j + x_j y_i + ... + x_N y_N) \\
&= (x_i y_i + x_j y_j) - (x_i y_j + x_j y_i) \\
&= x_i (y_i - y_j) - x_j (y_i - y_j) \\
&= (x_i - x_j)(y_i - y_j)
\end{aligned} \tag{4}$$

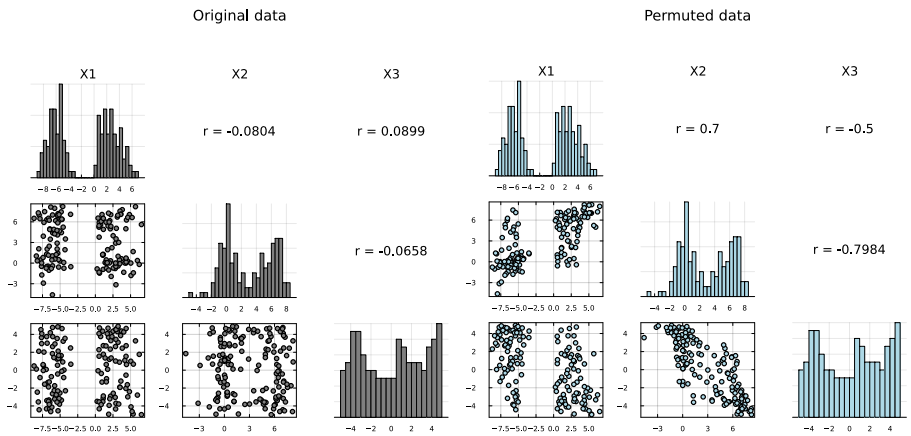

Figure 2: Permuting a random draw of 152 datapoints from three independent distributions: a Normal-Chisquare mixture ($X1$), a Laplace-Normal mixture ($X2$), and a Uniform distribution ($X3$). The left panel shows the original random draw, the right panel shows the data after permutation with our algorithm, with the target correlations set to 0.7, -0.5, and -0.8. Note that the marginal distributions, displayed on the diagonals, remain equal.

This allows us to precompute the change in the inner product as follows:

1. Pick a random index i.
2. For each $j \in 1...N$ compute $\Delta = (x_i - x_j)(y_i - y_j)$ as
3. Pick j such that the target inner product is most closely approximated (i.e., $|\tau_{xy} - \langle x, y \rangle - \Delta|$ is minimized)

4 Matrix extension

References

DiCiccio, C. J. and Romano, J. P. (2017). Robust permutation tests for correlation and regression coefficients. *Journal of the American Statistical Association*, 112(519):1211–1220.

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Schmidt, T. (2007). Coping with copulas. *Copulas - From theory to application in finance*, 3:1–34.